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Roll No .....

**ME-801(A)-GS**  
**B.E. VIII Semester**  
Examination, December 2020  
**Grading System (GS)**  
**Energy Management And Audit**  
(Elective - III)  
**Time : Three Hours**

**Maximum Marks : 70**

**Note :** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What do you mean by Energy Management? Explain it's importance.  
b) List the duties of a Energy Manager.
2. a) Discuss various types of energy audit instruments.  
b) What is FBC? Discuss it's salient features.
3. a) Explain the need of Energy Audit.  
b) How do you estimate energy cost?
4. a) What is Energy balance diagram? Explain its objectives.  
b) What are key elements of energy action planning?
5. a) Why force field analysis is needed in energy planning?  
How it is helpful?  
b) State brief about Energy Policy of India upto 2020.

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6.
  - a) Explain the concept of present worth method.
  - b) List the different types of depreciation methods.
7.
  - a) Discuss the areas of losses in electrical motors.
  - b) Define heat exchanger. State its types with neat sketches.
8. Answer any four of the following:
  - a) Assume yourself as energy manager, what step you would take for energy management for your organization.
  - b) Nowadays the concept of optimization in input energy requirement has evolved by substitution in fuel and energy. Explain.
  - c) Discuss the Sankey diagram and its use in material energy balance.
  - d) Discuss various methods for 'Building energy management'.
  - e) Explain thermal insulation in thermal systems. How thickness is determined?
  - f) How heat can be recovered from waste?

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